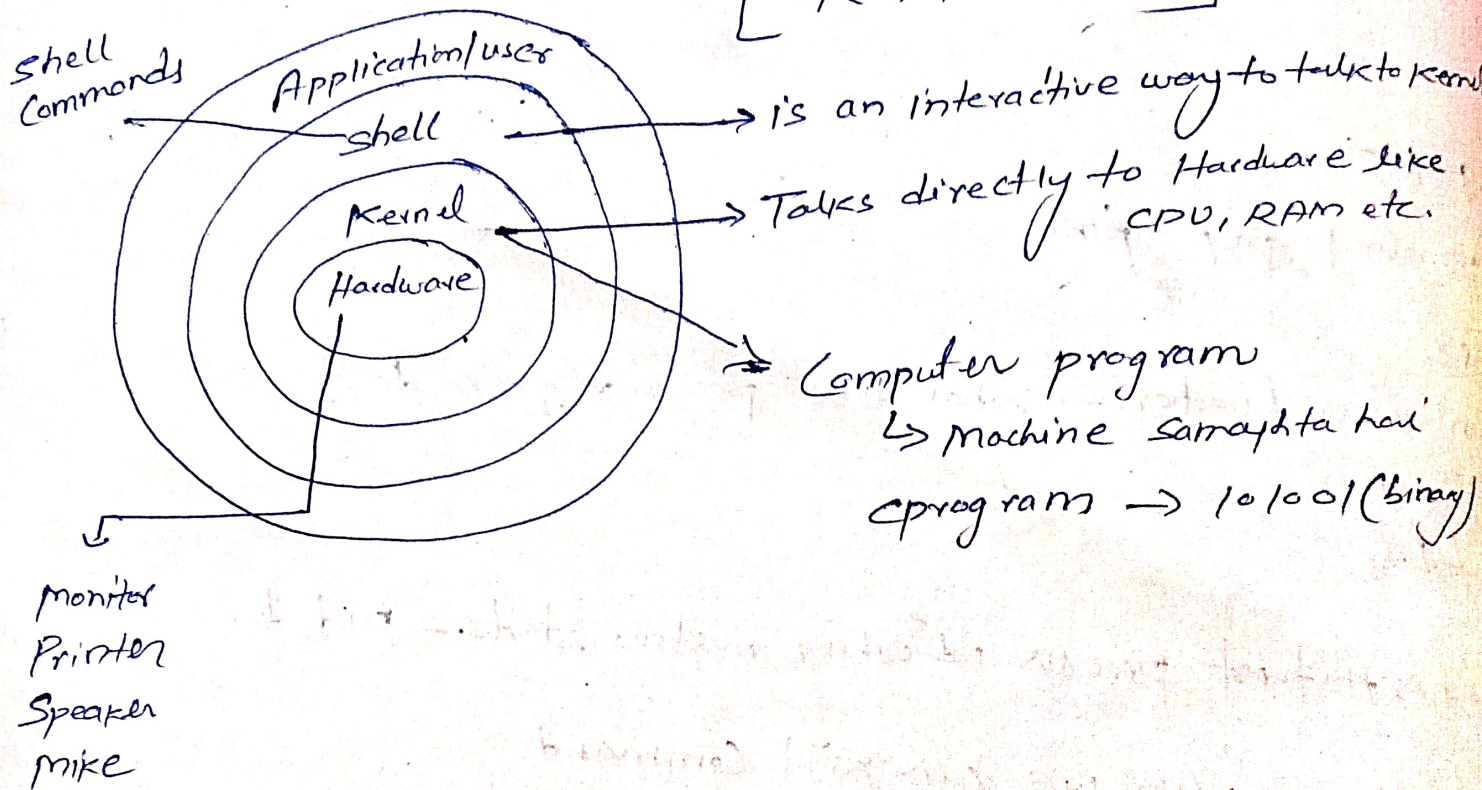


Linux DAY-2 Assignment

* Linux Architecture, processes and systemd *

* Linux Architecture *

A ⇒ Application
S ⇒ shell
K ⇒ Kernel



(userland)

* User Space * — The user space is a higher level portion of the OS where user facing applications and processes runs.

↳ Request serves from kernel

Note: [Directly interact nahi kar sakta hardware ke sath]

Example ⇒ ls → ask kernel → read disk → show file

* init / Systemd * Systemd is a service manager in linux which is responsible for booting OS.

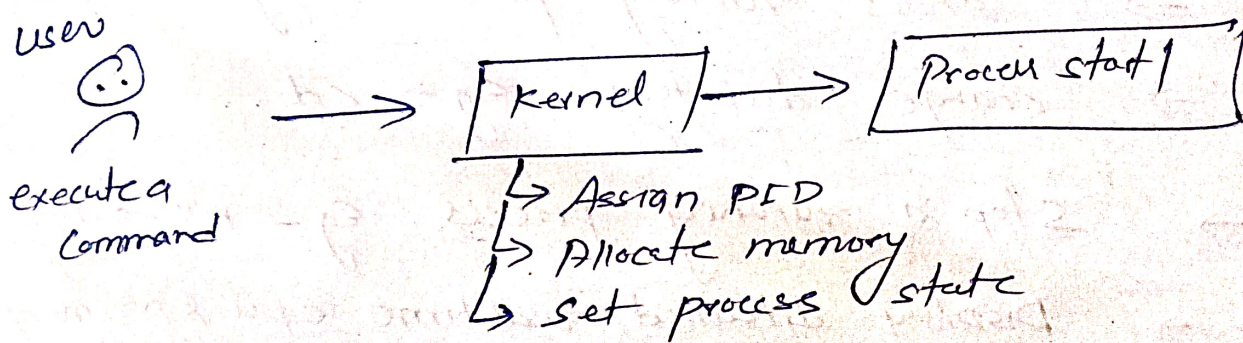
Note: Systemd is modern replacement for traditional init.

- The first process is started by kernel → PID 1.
- It start/stop/restart the services
- Handle logs as well.

Example - `systemctl start nginx` ↵
`systemctl stop nginx` ↵

Note! without systemd → system would not boot properly.

Process * A process is a program which is running in your system.



* Process states * Throughout lifecycle, process transitions through various states.

- i) Running (R) — using CPU right now
- ii) Sleeping (S) — waiting for H/W to run.
- iii) Stopped (T) — The process has been paused.
- iv) Zombie (Z) — finished or completed its task but its parent is crashed/died but are not aware about it.
- v) Idle → is has no other runnable tasks assigned to it and actively waiting for work.

* What systemd does and why it matters?

Ans! → Systemd is used to start, stop, restart the crashed services

→ Maintain logs

→ Handle boot targets.

* 5 linux commands *

i) pwd → shows current working directory → pwd

ii) ps → shows running processes eg - ps -e/aux

iii) cd → change directory eg → cd ..

iv) kill → stop a running process eg - kill -9 PID

v) Top / htop → Display detailed realtime cpu & memory usage